



NYU



Code
Available

ProCreate, Don't Reproduce! Propulsive Energy Diffusion for Creative Generation

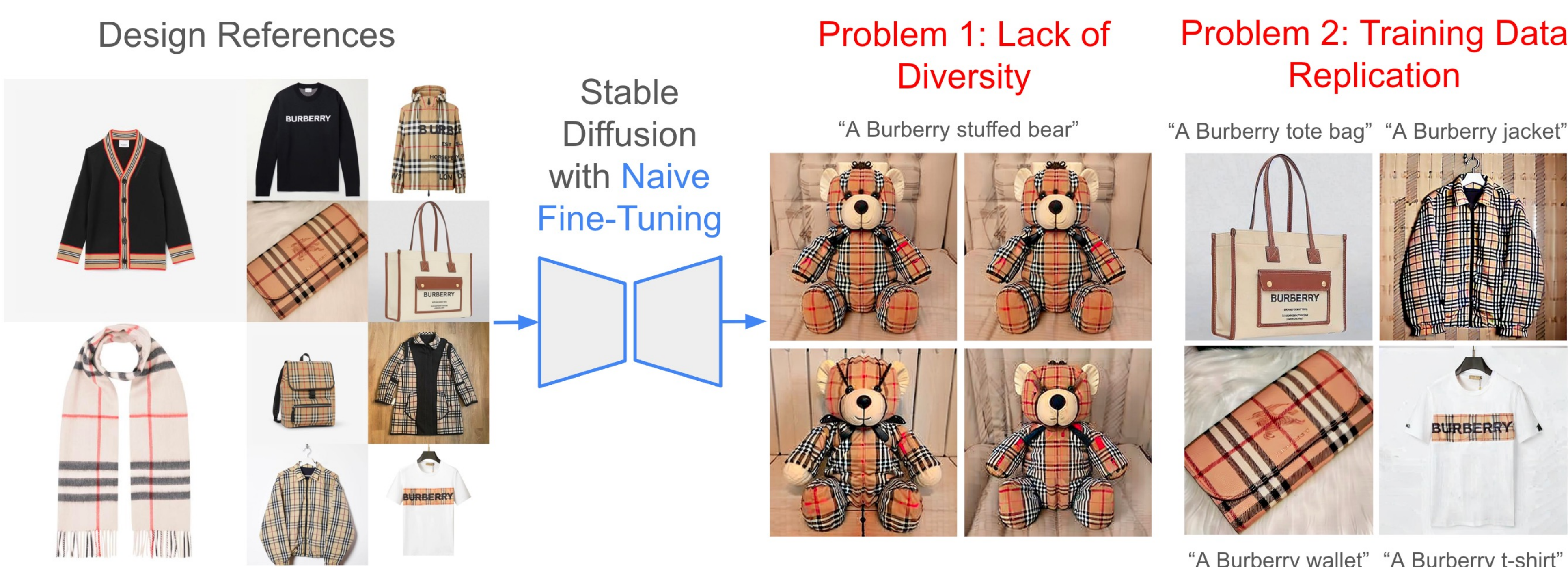
Jack Lu Ryan Teehan Mengye Ren



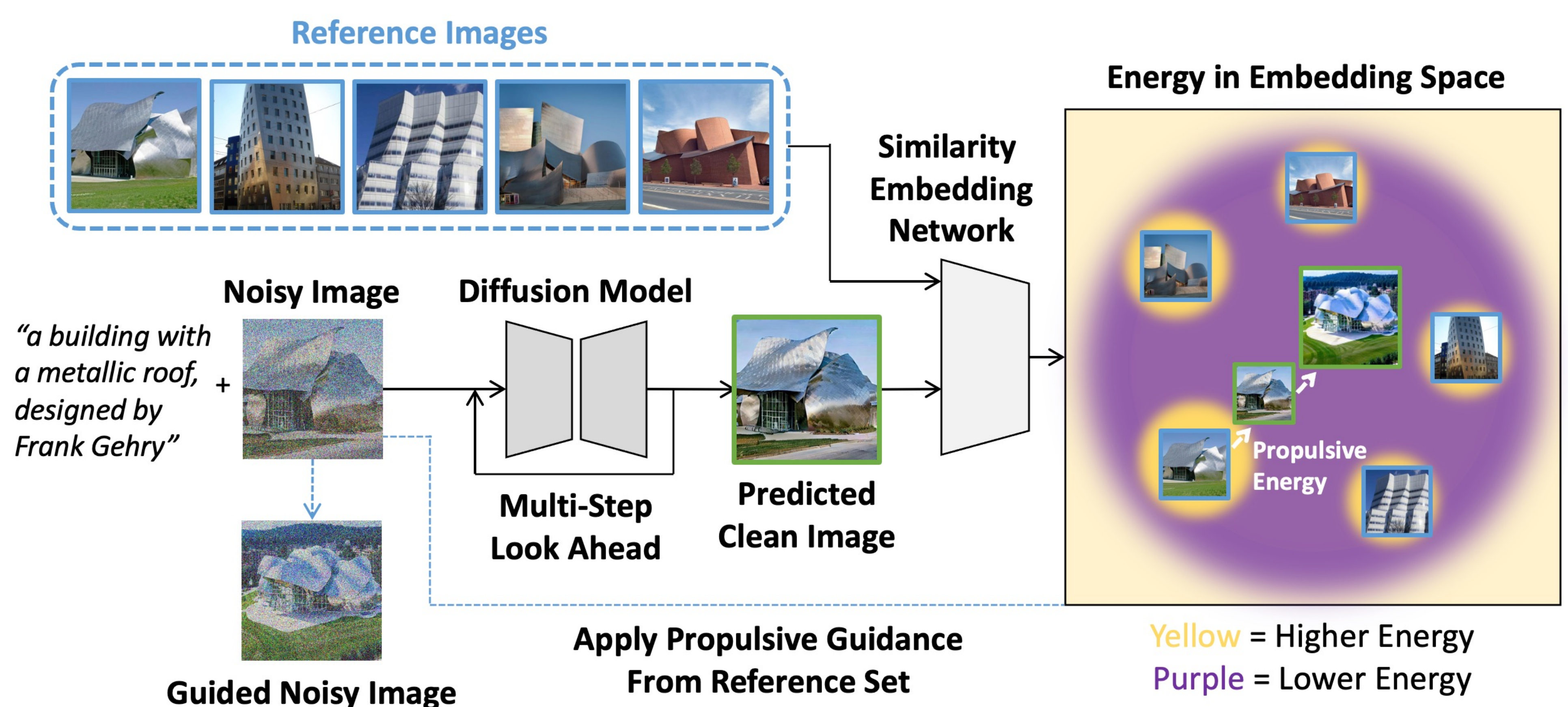
Motivation

Creativity: how to make diffusion models more creative by increasing the diversity of their samples?

Training data replication: how to prevent diffusion models from replicating content from their training sets?



Introducing ProCreate



ProCreate is a simple yet effective method where at each denoising step of a pre-trained/fine-tuned image diffusion model, we

1. Estimate a **Predicted Clean Image** with **Multi-Step Look Ahead**, i.e., multiple DDIM sampling steps
2. Embed the **Predicted Clean Image** and the **Reference Images** (e.g., training data) with a **Similarity Embedding Network**
3. Match the **Predicted Clean Image** with its closest reference in the embedding space
4. Use their distance's gradient to push the **Noisy Image** away from the closest reference

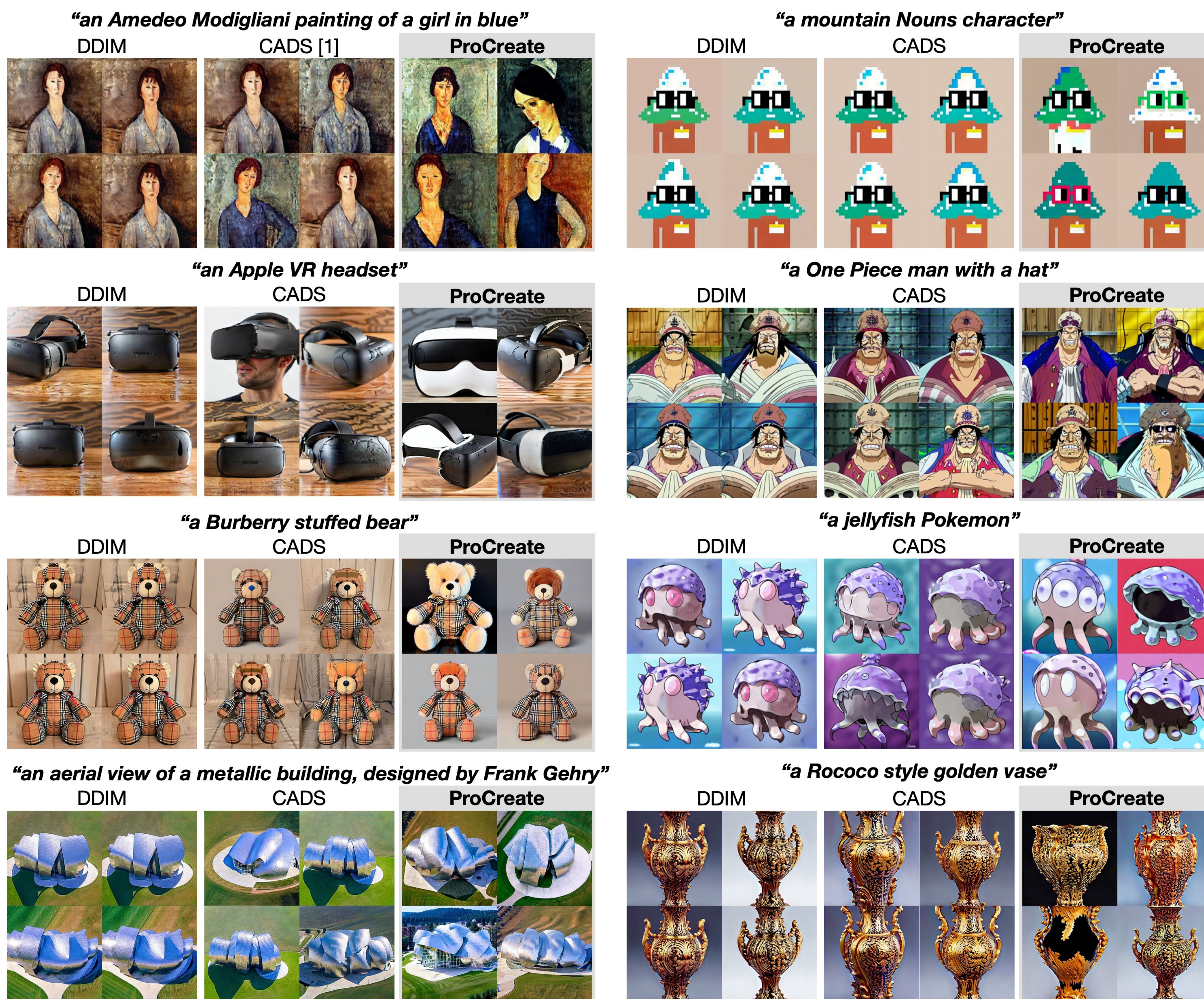
New Dataset: Few-Shot Creative Generation 8 (FSCG-8)

A few-shot dataset with 8 categories of 50 image-caption pairs each. Each category contains images that share properties like style, texture, shape, etc.



Application 1: Few-Shot Creative Generation

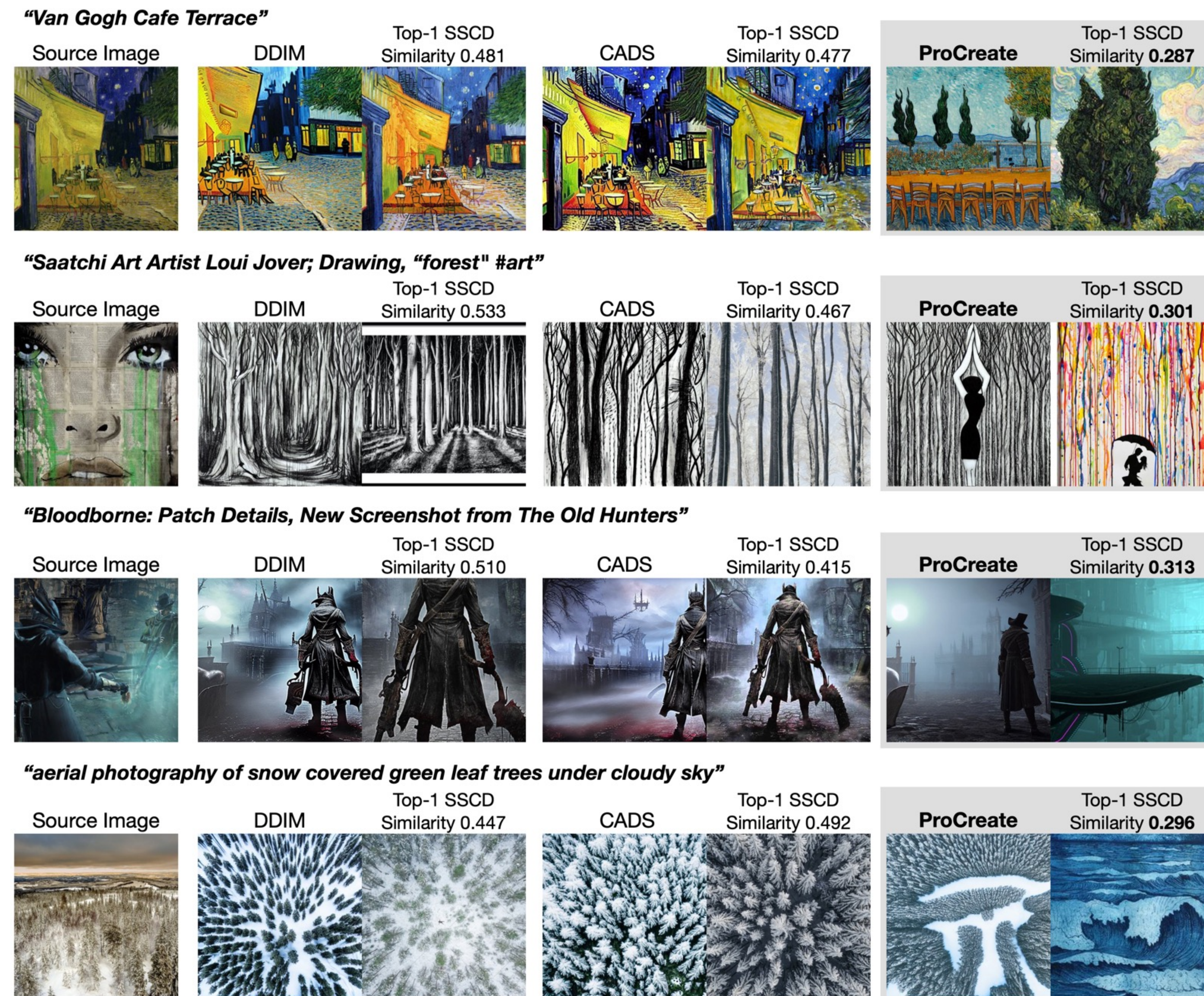
Fine-tune a separate Stable Diffusion model on each category of **FSCG-8**. Generate new, diverse samples from each category with **ProCreate**.



[1] S. Sadat et al. CADs: Unleashing the Diversity of Diffusion Models through Condition-Annealed Sampling. ICLR 2024.

Application 2: Preventing Training Data Replication

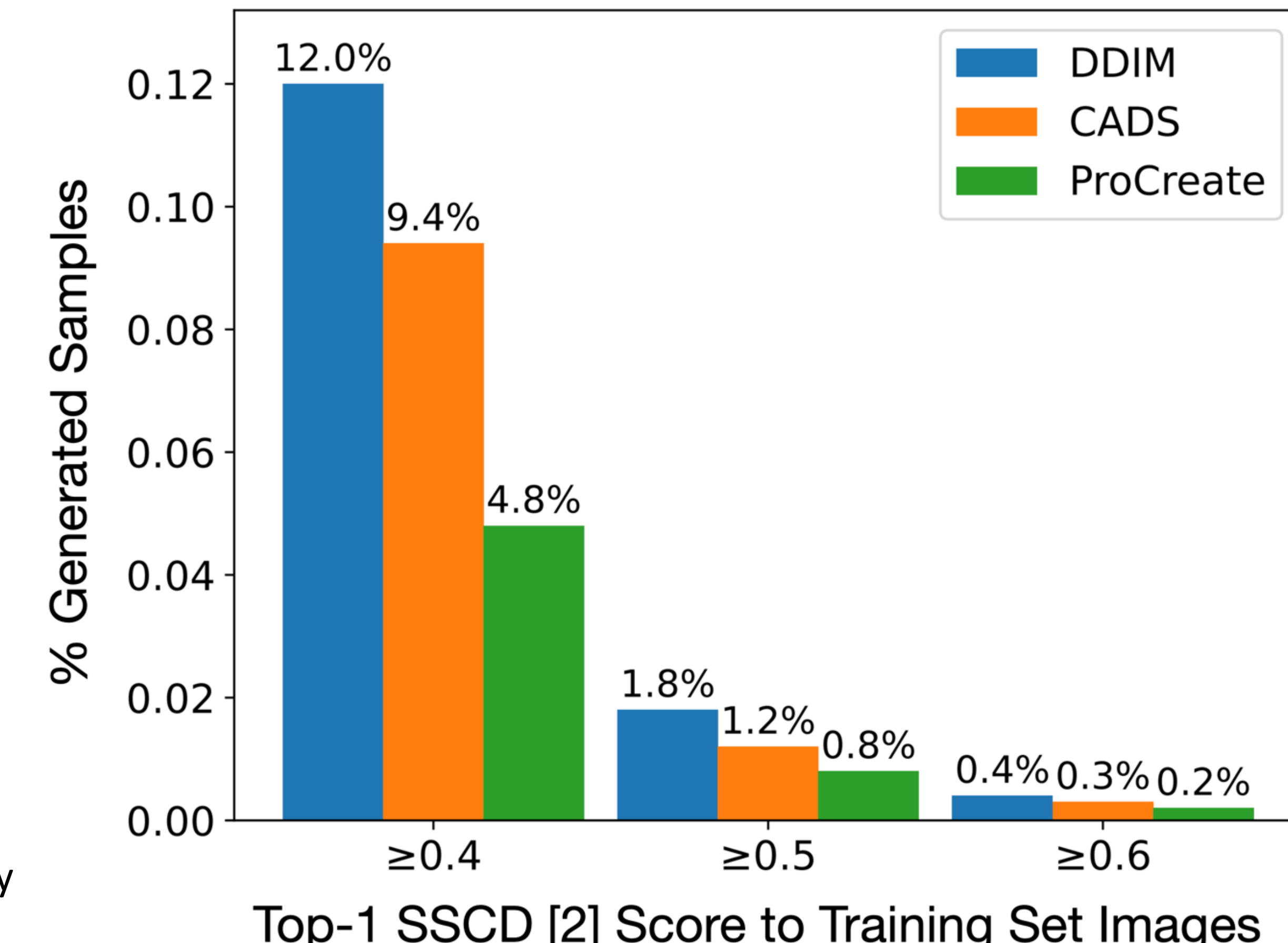
Sample from Stable Diffusion with captions from its pre-training dataset (LAION5B). Use **ProCreate** to not replicate LAION5B images in generated samples.



ProCreate reduces the replicate rate of generated samples →

ProCreate improves distribution matching metrics ↓

	FID	KID
DDIM	1.38	0.051
CADS	1.60	0.065
ProCreate	1.14	0.038



[2] E. Pizzi et al. A Self-Supervised Descriptor for Image Copy Detection. CVPR 2022.